LAB #2 – Array Operations in MATLAB

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Purpose:

- MATLAB and simple arrays.

Before the lab:

- Read slides and chapters 2 and 3 (first half) of the textbook.

During the lab, please do the questions by order.

1. With MATLAB, do the problems 4 to 8 in your textbook chapter 2 (section 2.11) and problems 3 to 5 of chapter 3 (section 3.9).
2. In your textbook, look at the more advanced problems after number 20 (section 2.11 - chapter 2), select 22, 23, 24, 37 and 38 from those and do them with MATLAB. Write the question index when you submit the report.
3. Close MATLAB.
4. **Tracing Exercises:** Evaluate what will be the values of **p**, **q**, and **r** after these MATLAB instructions will be executed (**Do these by hand: Do not use MATLAB!**). Must show all your calculation procedures.

```
n1 = [1:5:9];
n2 = [10 14 4];
n3 = [length(n1) 0 0];
p = [n1*3, n2, n3-n2];
n4 = [10, 16, 3.3, 6];
m1 = [1 2 3 4; 5 6 7 8];
m2 = [m1;n4];
q = ones (4,3) + m2';
n5 = [10:20:60];
m3 = [eye(3) .* 10; n5];
s = [m3(3,2:3), m3(1:2,3)'];
s(3) = 12;
r = [m3(3,:), s];
```

5. Now, finish the following steps.
   A. Open MATLAB.
   B. In the command window, enter **edit z1**. The script editor should appear.
   C. In the script editor, enter the following script. Please assign your first name to variable ***firstname*** in the file.

```
firstname = 'Your first name here';
s1 = 10;
s2 = 21;
s3 = s1 + s2;
s3
```

   D. Save your script. It will be saved under the name **z1.m**.

E. In the command window, enter **edit z2**.

F. In the script editor, enter the following script:

```
firstname = 'Your first name';
x1 = 200;
x2 = 300;
x3 = (x1 * x2) + 300;
x3
```

G. Save your script. It will be saved under the name **z2.m**.

H. You're done for this lab. Close MATLAB and save z1.m and z2.m for your lab report submission purpose.

Lab Report Submission Requirement:

1. Submit Lab #2 based on the given templates on the Blackboard. Please post all the screenshots of the inputs and running outputs of each step on the lab manual.
2. Attach z1.m and z2.m with your submission.
3. Please submit your report on the blackboard by the due time.